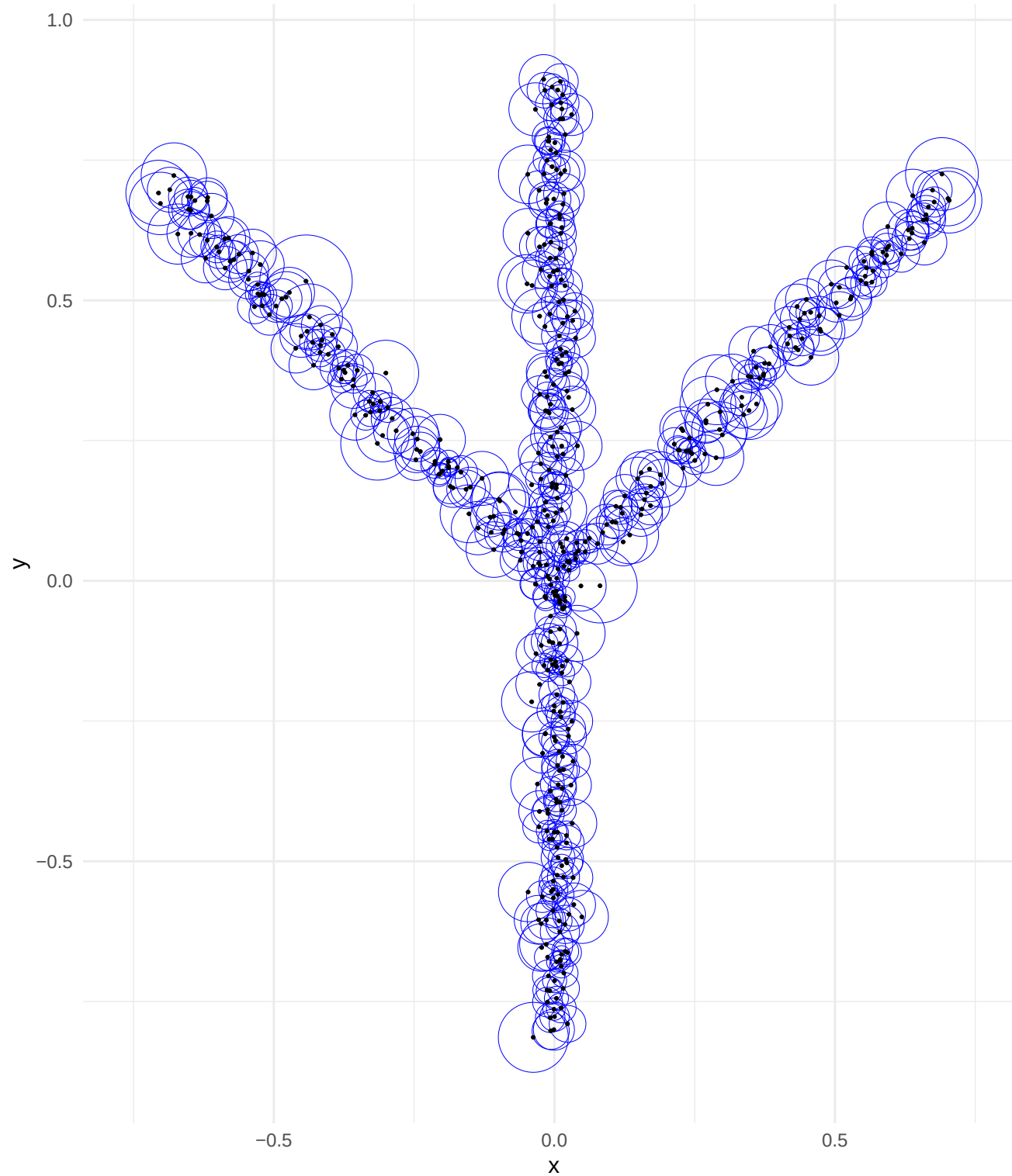
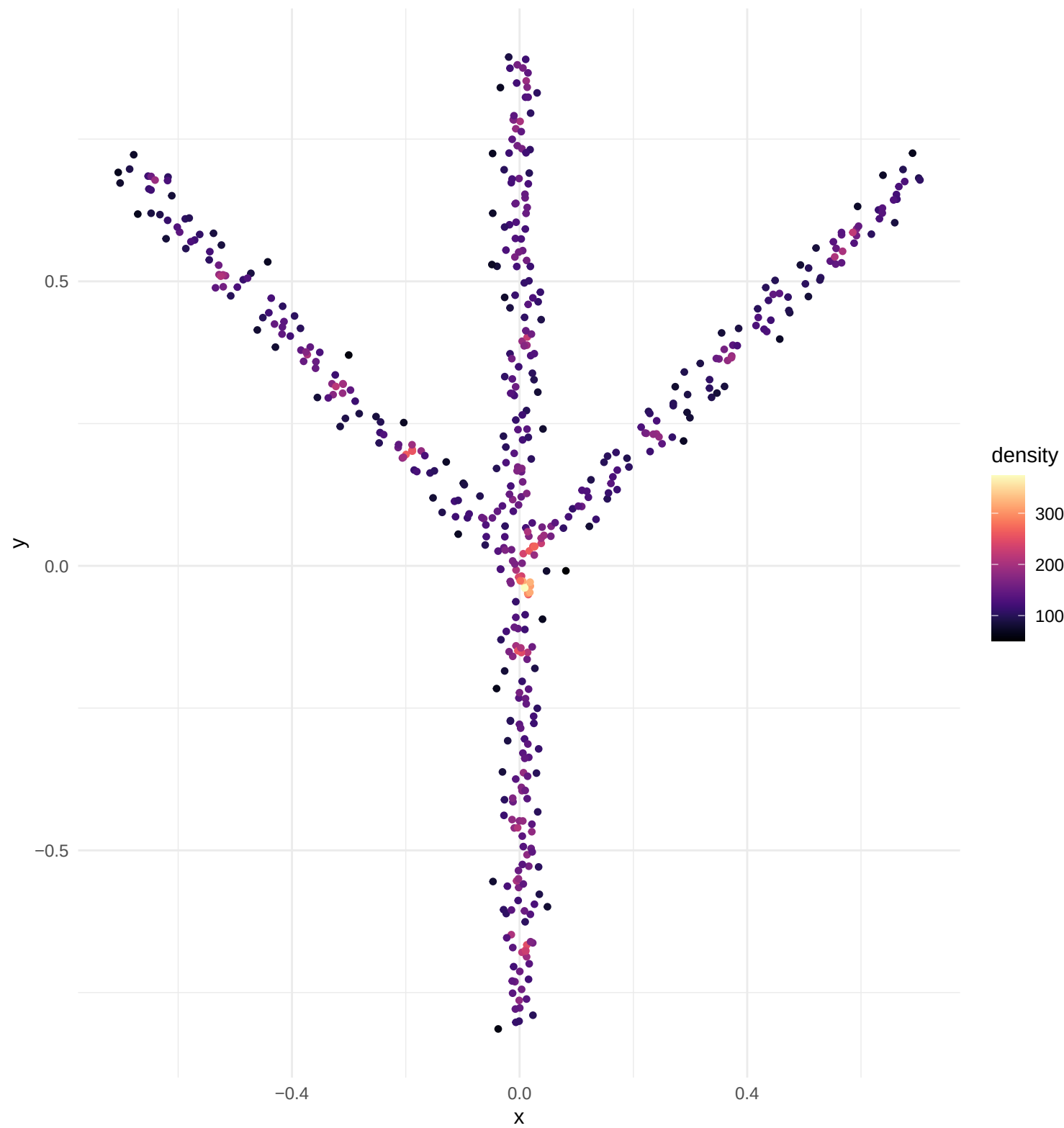


All Adaptive Spheres

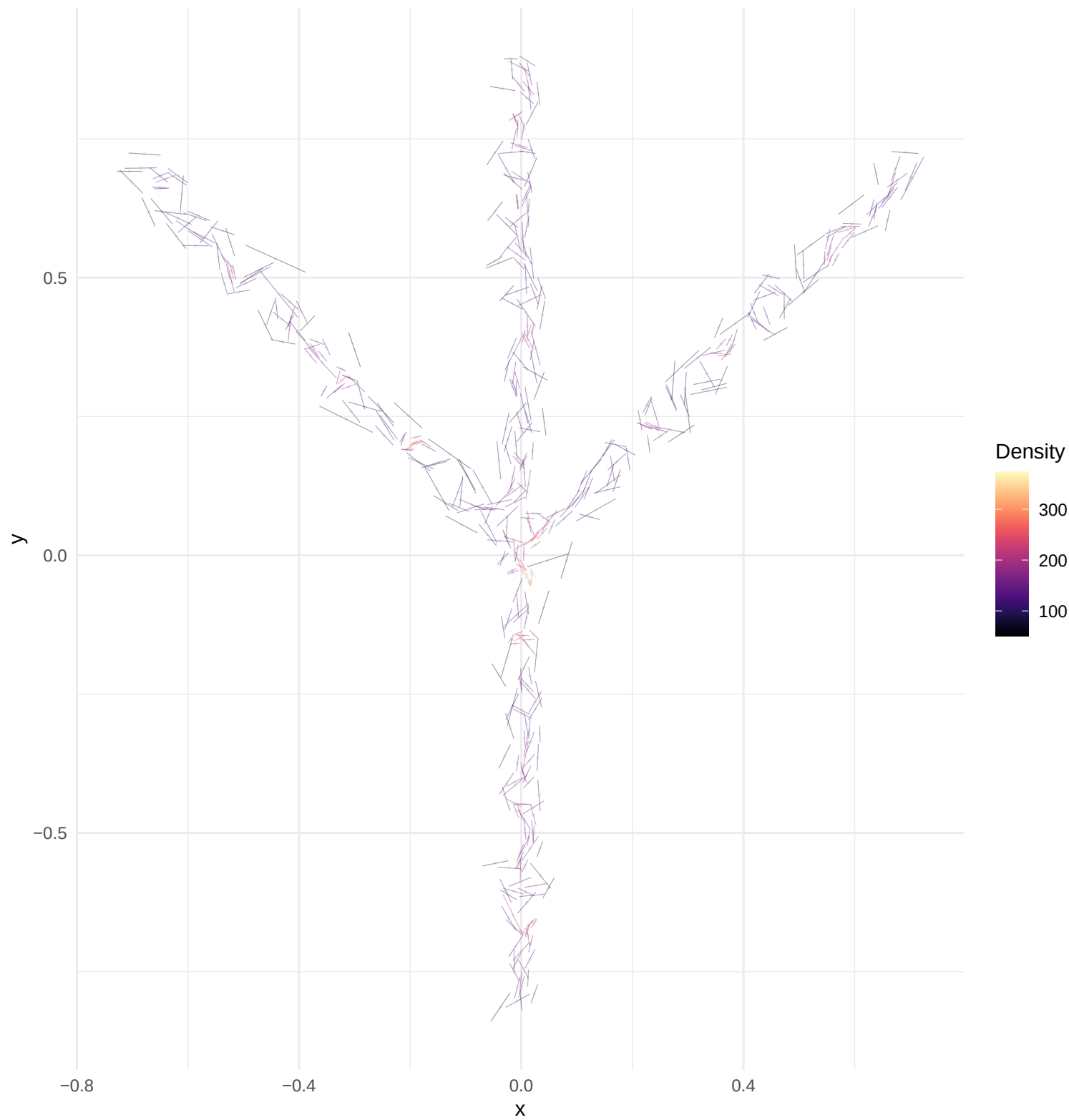
Visualizing local neighborhood size (radius) for every point.



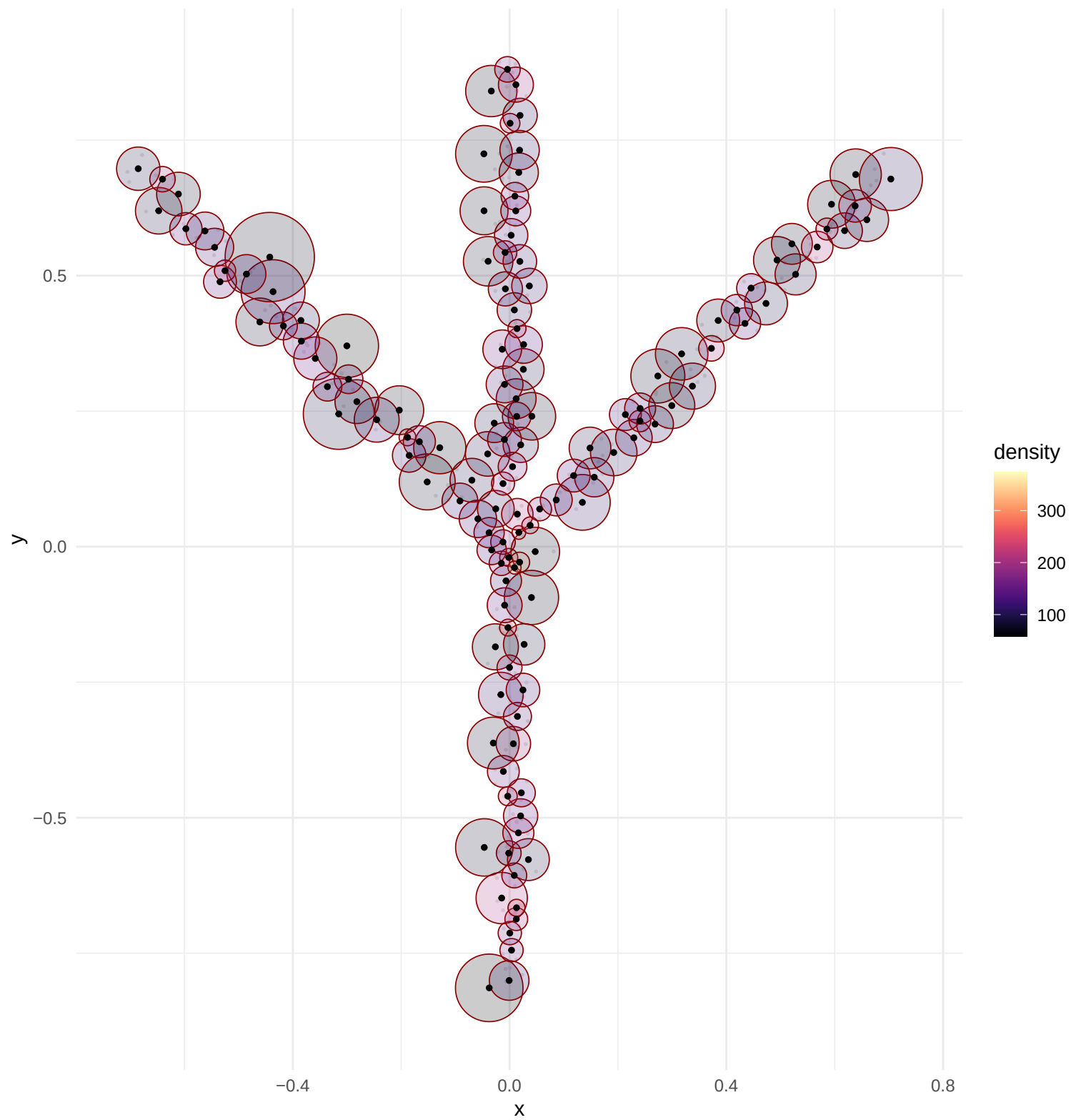
Point Density Distribution



Global Tangent Field (PC1)

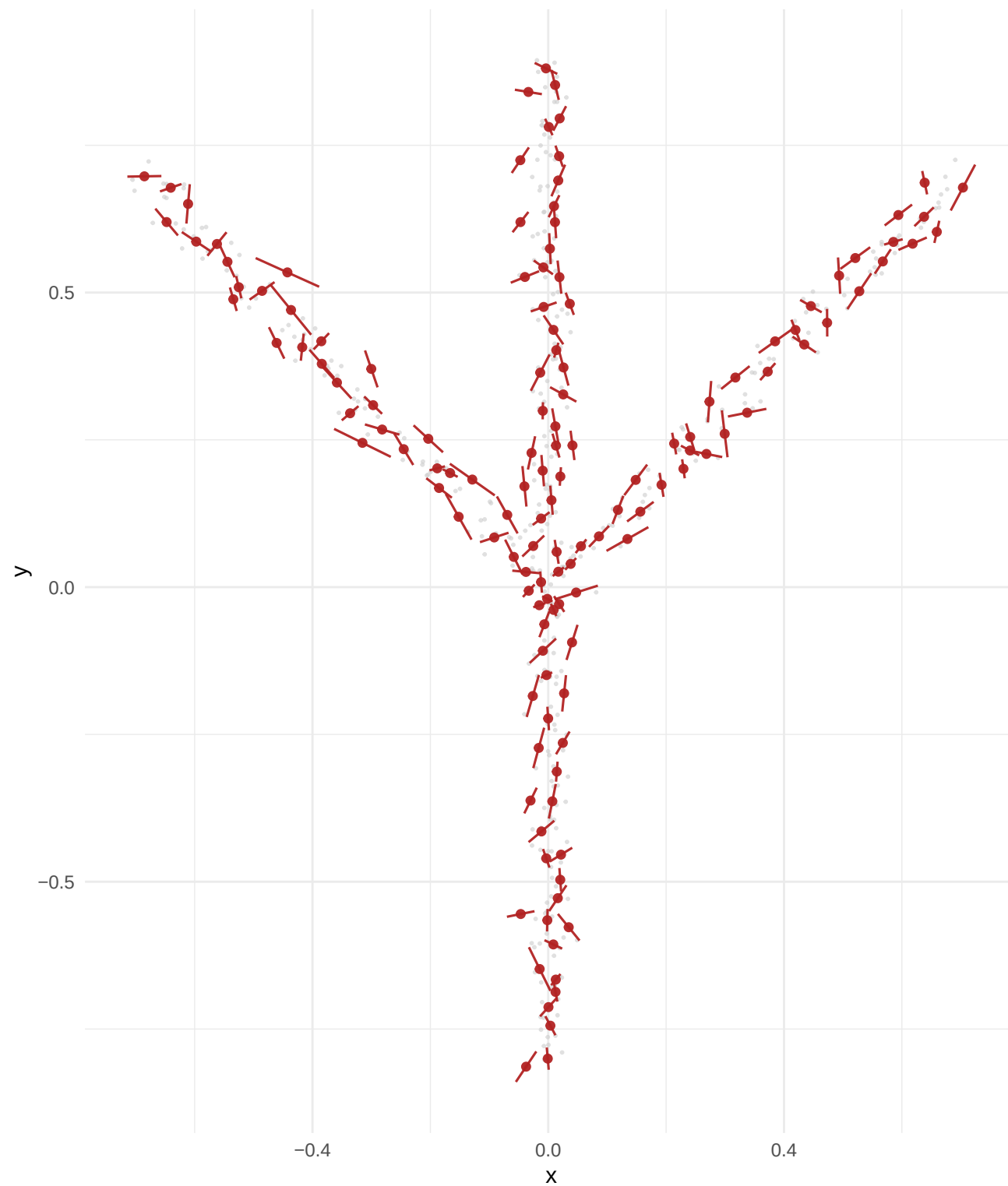


Atlas Selection (Anchors)



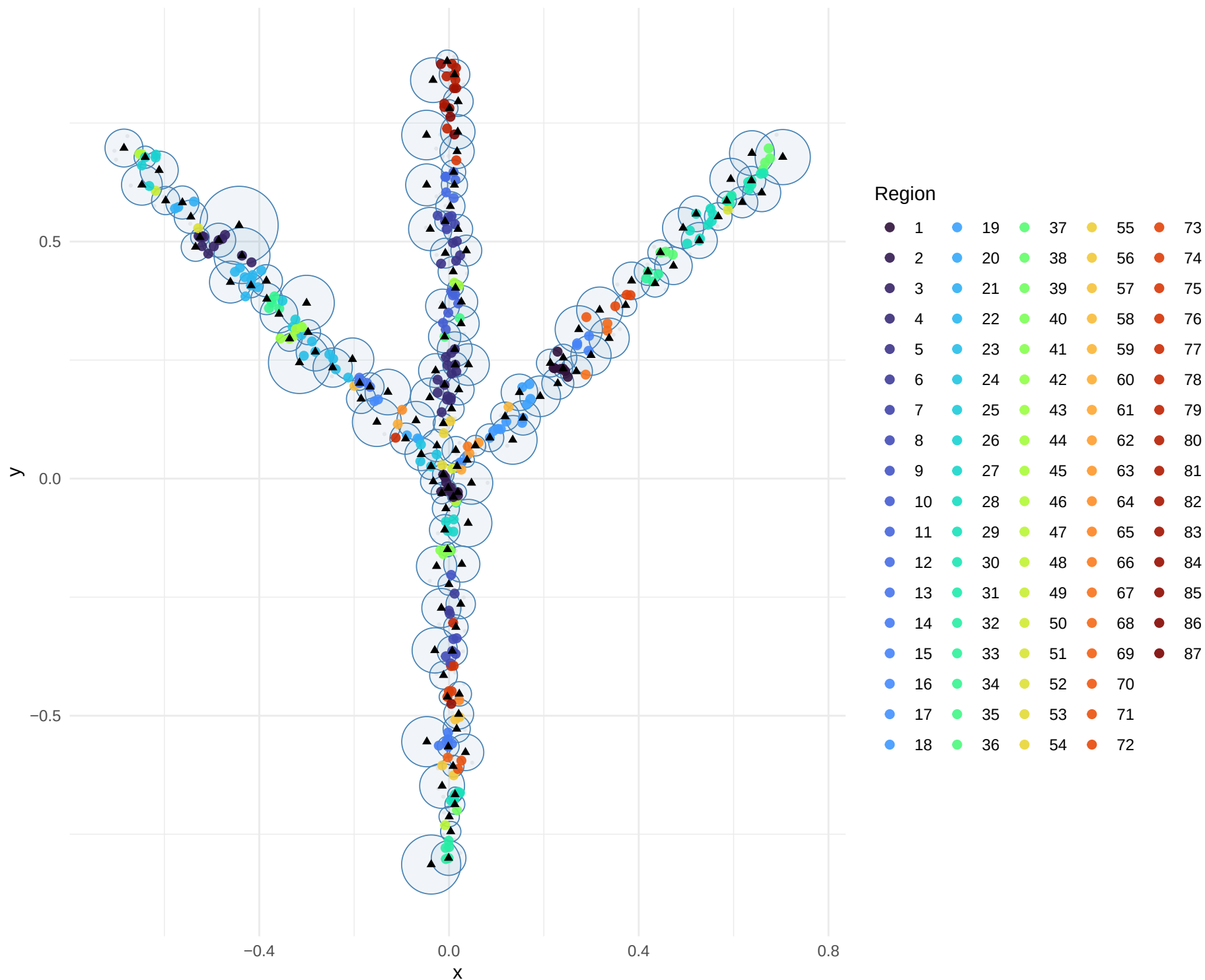
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



Intersection Regions (2+ anchors)

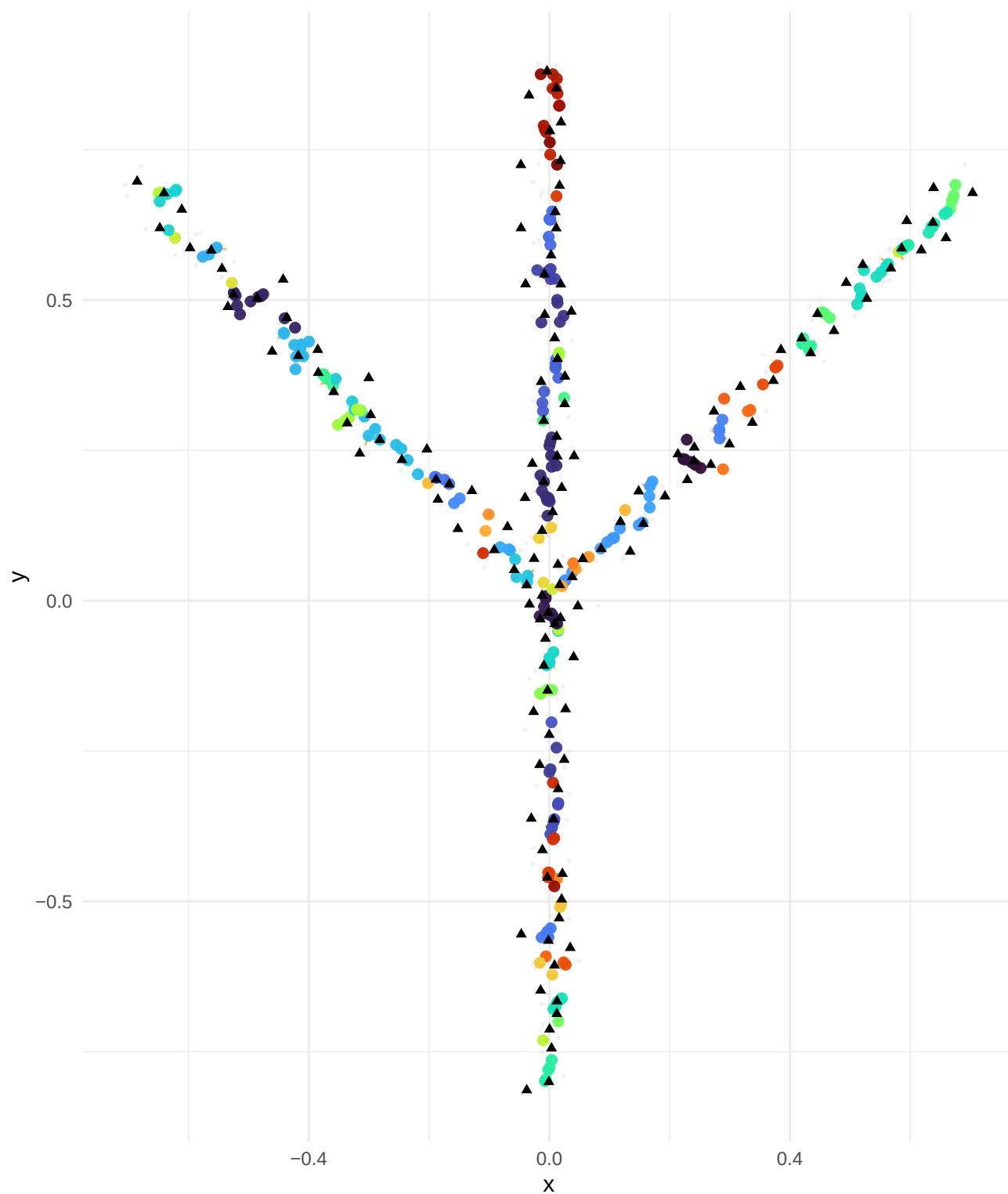
Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01

Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors

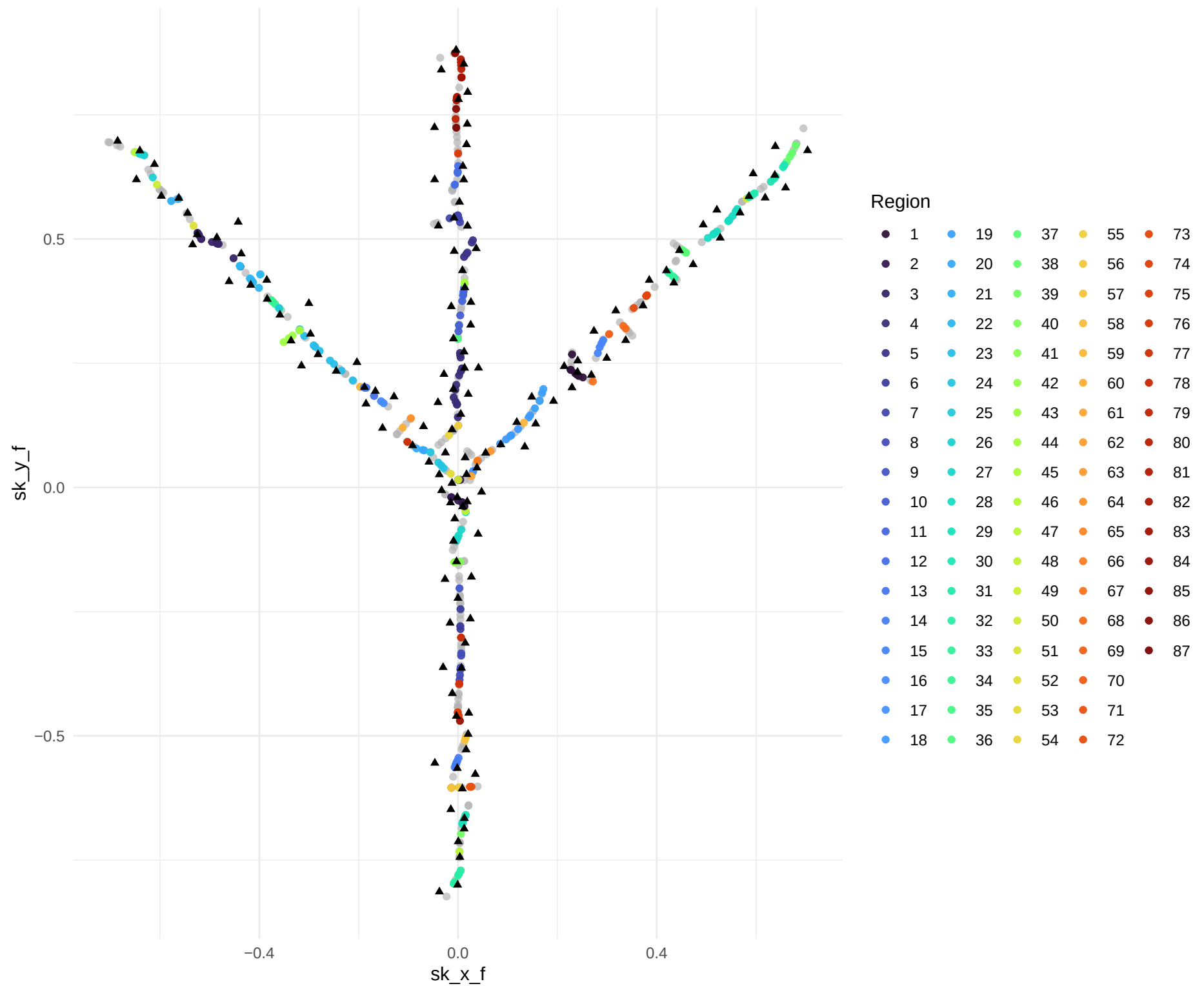


Region

1	19	37	55	73
2	20	38	56	74
3	21	39	57	75
4	22	40	58	76
5	23	41	59	77
6	24	42	60	78
7	25	43	61	79
8	26	44	62	80
9	27	45	63	81
10	28	46	64	82
11	29	47	65	83
12	30	48	66	84
13	31	49	67	85
14	32	50	68	86
15	33	51	69	87
16	34	52	70	
17	35	53	71	
18	36	54	72	

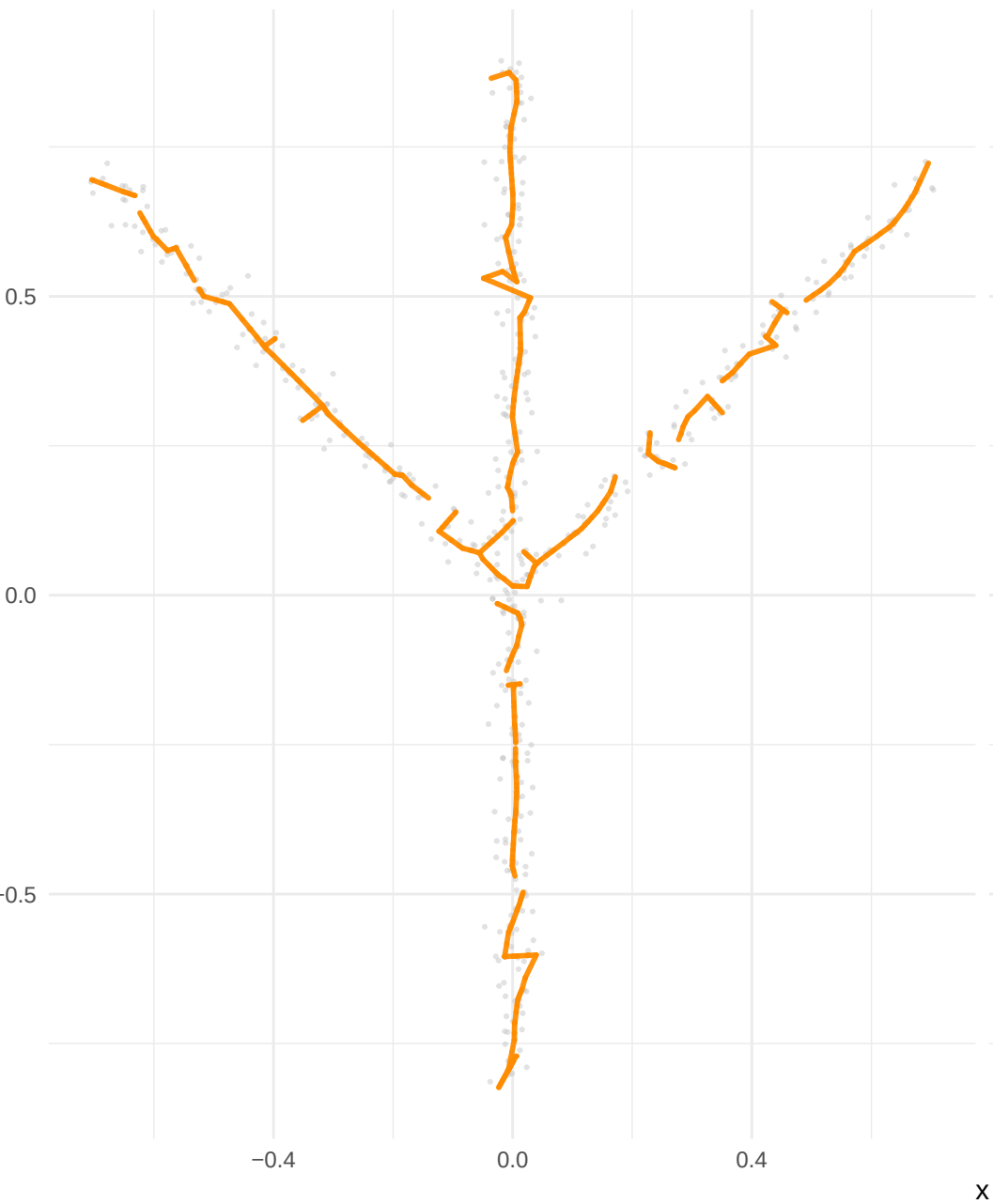
Final Stitched Backbone (Converged)

Final stitched points

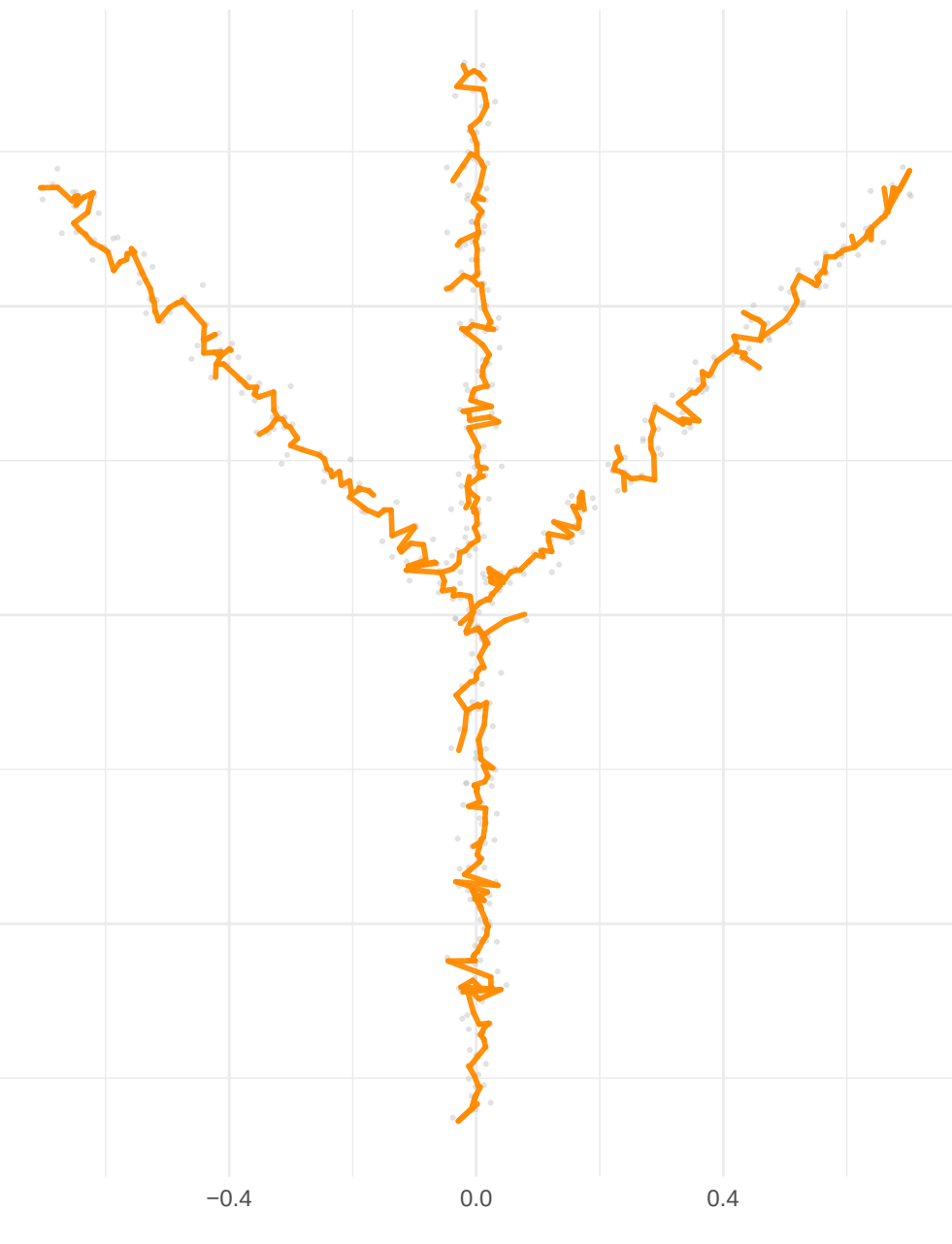


Backbone Edges

Final

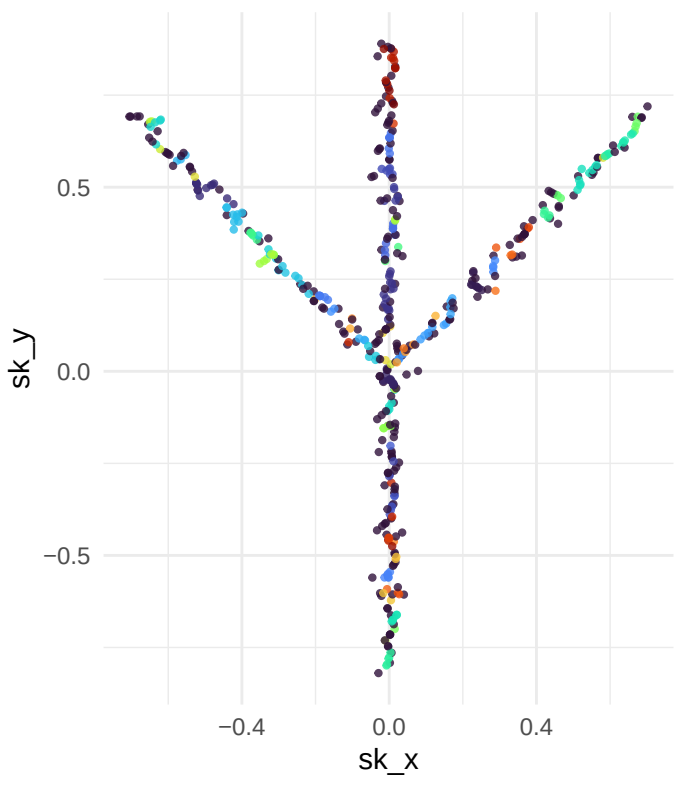


Iteration 1

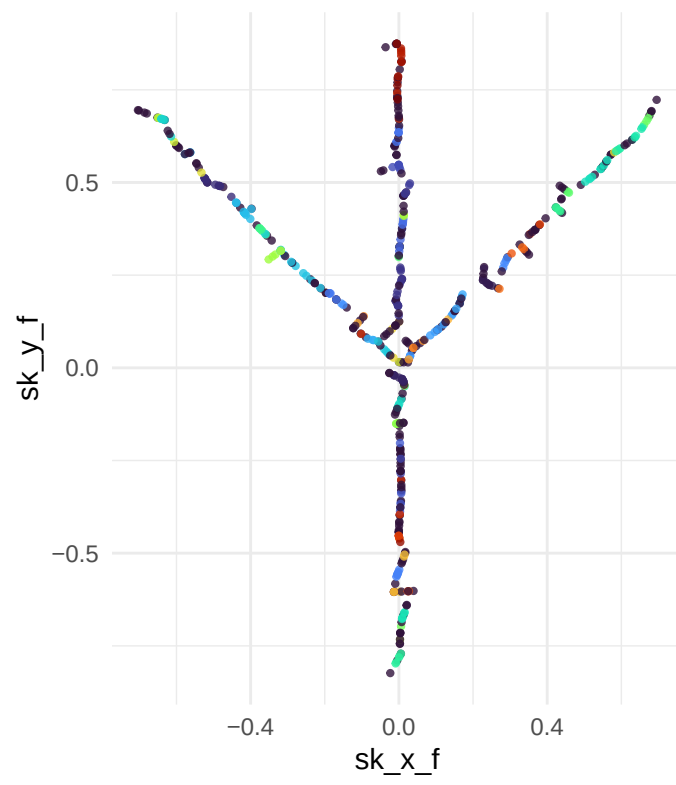


Convergence: iteration 1 vs final

Iteration 1

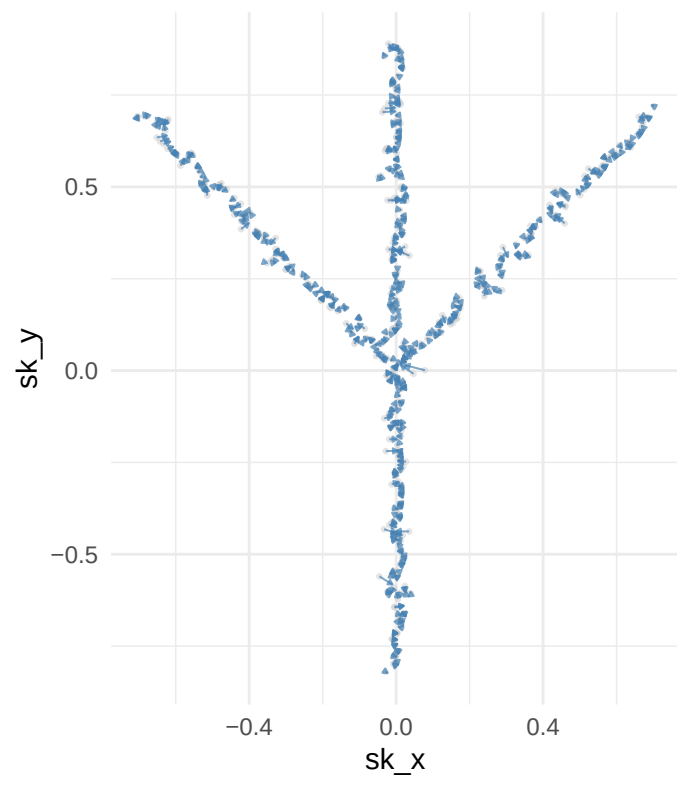


Converged



Displacement iter1 -> final

Points moved > 1e-6: 500 / 500



Parameters: $k_min = 5$, $g_threshold = 3.0$, $o_max = 0.3$, $o_min = 0.10$, $stability = 0.90$, $scale_factor = 2.5$, $max_iter = 50$, $conv_tol = 0.01$